



Anti-mosquito clothing

A new study shows that graphene sheets can block the signals mosquitos use to identify a meal. <https://digit.in/oct19-55>



Pesticides harm birds

A study show that the decline of migrating birds could be partly due to pesticides. <https://digit.in/oct19-56>

WHY DO WE SLEEP?

While there are some theories, we don't really know

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umans spend almost one third of their lives asleep in bed. At least, they need to if they want to live a healthy

life. All animals need to sleep, and yet scientists bafflingly don't yet know exactly why. There are some animals that sleep very very little. Marine mammals and migrating birds are examples of these. The marine mammals need to keep surfacing for air, and the birds may need to keep flying through the day and night. The dolphin and the Alpine swift both sleep with only half a brain, and keep the other half occupied with the business of moving and watching out for predators. The ostrich sleeps standing up, with its eyes open and watchful for predators. However, in captivity, dolphins are known to sleep entirely, with a reflex action keeping their blowholes above the surface. There is no animal that does away with sleep entirely. Sleep

deprivation can lead to all kinds of negative consequences to the health.

In fact, one of the ways that scientists have explored the function of sleep is to check what happens when humans do not go to sleep. The blood pressure shoots up, they cannot concentrate or remember things, their coordination and balance goes for a whack. Long term effects include obesity, diabetes and a weakened immune system. All of us know from personal experience that a good long period of sleep makes us more alert and energetic. That, however, is only the beginning of an explanation. Why we need food for example, is very clear cut. It is an energy source. Why we need to sleep though, is not as clear. Scientists have some theories to explain the function of sleep though.

EVOLUTIONARY THEORY

According to the evolutionary or adaptive theory, sleep emerged as a way for animals to stay safe at night. By going

to a hiding place and stopping all signs of movement, a creature is less exposed to predators, and does not do any activities that are more dangerous at night. For example, foraging for food in low-visibility conditions may make creatures vulnerable to scratches and falls. Some scientists believe that for most animals, working in the day and resting at night has become a part of the neurochemistry of the brain. Essentially, according to this theory, the animals that did sleep had an evolutionary advantage over animals that did not, and this amplified through time. The animals at the top of the food chain spend a lot more time sleeping as against the animals towards the bottom. For example, a bear may sleep for 12 to 15 hours a day, and lions sleep up to 20 hours a day. A deer on the other hand is lucky if it manages 4 hours of shut eye. Although the evolutionary theory emerged from a comparative study of how long animals sleep, staying alert for predators is